
How to Find Emerging Titans: A Beginner's Course

As of Sep 27, 2026

This course assumes you're starting from zero. Every term is defined the first time it appears. Where research data exists, the actual numbers and source are given. Where it doesn't, that's stated, and estimates and opinions are labeled.

Lesson 1: What is an "emerging titan"?

An emerging titan is a small company that grows into one of the most important companies in its industry, and eventually the economy. Think of NVIDIA, Amazon, Netflix, Microsoft, or Apple before they were famous.

How rare they are. J.P. Morgan studied the Russell 3000 (the 3,000 largest US companies) from 1980 to 2020:

- About **40%** of stocks suffered a "catastrophic loss": a drop of 70% or more that never fully recovered.
- About **two-thirds** did worse than the index, and about **40%** lost money outright.
- Only about **10%** were big winners, and those produced most of the index's gains.

Hendrik Bessembinder (2018) found that just **4%** of US-listed companies accounted for all of the stock market's net wealth creation from 1926 to 2016.

Failure rates are often overestimated. People usually guess about 90% of dot-com companies failed. A University of Maryland study found **48%** of a random sample of 1999 venture-funded dot-coms were still in business five years later. Roughly half didn't survive, and the survivors mostly didn't become titans.

What researchers found big winners had in common. Christopher Mayer studied 365 stocks that rose 100-fold between 1962 and 2014. He identified:

- High returns on capital
- A long runway for growth
- The ability to reinvest profits at high returns
- Small starting size
- A durable competitive advantage

How far big winners fell along the way:

- **The best performers:** among the 100 best stock-decade performances, the average decline during the winning decade was over 30%, and over 50% in the decade before the breakout (Bessembinder).
- **Recent 10-baggers:** US stocks that rose tenfold over the last five years had a median worst decline of 56%; six fell 80% or more at some point.
- **Famous examples:** Amazon fell about 94% after the dot-com crash, Netflix roughly 75-80% in 2011, and NVIDIA was cut in half more than once.

Survivorship bias: the 40% catastrophic-loss figure shows that falling far is also common among losers. A big drop doesn't signal a future titan. Studying only winners makes their path look safer than it was.

Lesson 2: The math of size

Market capitalization (market cap) = share price x number of shares. It's the price to buy the whole company. 10 million shares at \$50 = \$500 million.

For your money to grow 100 times (a "100-bagger"), the company's value usually has to grow roughly 100 times too. Buybacks help you; dilution hurts you.

Starting value	Value after 100x	How many companies are that big?
\$500 million	\$50 billion	Several hundred worldwide (estimate)
\$5 billion	\$500 billion	Roughly 30 worldwide (estimate)
\$45 billion	\$4.5 trillion	None today

Rocket Lab (about \$45 billion) can't realistically be a 100-bagger from here. Its big multiplier already happened: it traded around \$4 a share in 2023.

Starting size data: in Kevin Martelli's study (cited in Mayer's book), **68%** of multibagger stocks traded below a \$300 million market cap at their low. AMD was worth roughly \$1.5-2 billion in early 2016, and NVIDIA roughly \$10-18 billion during 2015 (approximations).

The \$500 million to \$5 billion range is a personal guideline for where 100x is mathematically realistic. Martelli's data suggests many big winners started smaller.

Opinion: by the time a stock is widely famous, much of its early growth has usually happened.

Lesson 3: Waves (the change bigger than any company)

Industry: a group of companies selling similar things. **Total addressable market (TAM):** what customers could spend if everyone who needs it bought it.

A great company in a small market has a lower ceiling than a decent company in an exploding one. In one study of stocks that rose tenfold from 2009 to 2023, technology produced the most, with industrials and consumer discretionary close behind; real estate, energy, and utilities produced the fewest.

The S-curve. New technologies tend to spread in three stages:

1. **Slow start:** only early adopters; expensive and clunky.
2. **Explosion:** prices drop, quality rises, adoption climbs fast.
3. **Maturity:** almost everyone has it; growth slows.

Smartphones: about **35%** of US adults owned one in 2011, about **81%** by 2019 (Pew Research). The 10%/80% stage boundaries often quoted are rules of thumb, not laws.

The best time to spot a titan is late in stage 1 or early in stage 2: the technology clearly works, but most people don't use it yet.

Lesson 4: Second-order thinking (picks and shovels)

First-order: what's the obvious winner? **Second-order:** what does the obvious winner need that's in short supply?

Gold Rush: no reliable count exists of how many miners succeeded, but accounts agree striking it rich was uncommon, while some lasting fortunes came from supplying miners (Levi Strauss is the famous example).

The modern chain:

1. AI needs chips, so everyone bought NVIDIA.
2. AI data centers need enormous amounts of electricity.

3. Delivering it needs transformers, switchgear, and transmission lines, already in short supply.

At step 3: commercial customers, including data centers, went from about 3% of Powell's backlog in 2021 to about 40%; Hammond's backlog was up 122% year over year at the end of 2025.

Rule: when a trend becomes obvious, look one or two steps behind it for the bottleneck.

Lesson 5: Six signals an industry is about to boom

1. **Big buyers commit money before supply exists.** In 2026, Amazon guided to about \$220 billion of capital spending and Meta to \$130-145 billion, largely for AI data centers. Strong evidence of real demand, though buyers can overcommit and cancel.
2. **Lead times stretch.** Lead time = wait between ordering and receiving. Reported lead times for large power transformers stretched to roughly two to four years.
3. **Costs fall fast.** IRENA reported utility-scale solar electricity costs fell about 85% from 2010 to 2020.
4. **Adoption is still low:** the edge of the S-curve's steep section.
5. **The first pure-play turns profitable.** A pure-play does only that one thing. Its profit is evidence the model can work, not proof for everyone.
6. **The stocks haven't already spiked.** No exact threshold for "already discovered"; it's a judgment call.

Lesson 6: The nine traits of an emerging titan

1. **Rides a wave bigger than itself.** Check: is the industry expected to outgrow the economy for 10+ years?
2. **A real, selling product with accelerating growth.** Revenue = all money from sales before costs. Accelerating = growth speeding up (Hammond: 11%, 14%, then 38%). In the 2009-2023 tenfold-stock study, median revenue growth was 11.1% a year and EPS growth 20.0%: steady growth plus rising profitability, not extreme growth.
3. **A moat that strengthens with size.** Moat = what keeps competitors from stealing customers. Types: switching costs (utilities' hardware approvals likely create them for PLPC; an inference, not quantified by the company), network effects, ecosystem lock-in (NVIDIA's CUDA), scale (Amazon's warehouses). Warning: a capacity moat tends to vanish when competitors build factories.
4. **Rising gross margins.** Gross profit = revenue minus direct production cost; gross margin = that as a % of revenue (\$100 sale, \$66 cost = 34%). Rising while growing suggests pricing power. Red flag: American Superconductor's drop from over 30% to 26.3%.
5. **A long-term leader with real ownership.** Mayer's research noted a large owner or entrepreneurial founder was often involved. Look past the CEO: PLPC's CEO owns well under 1%, but the March 2026 proxy shows Rob Ruhlman at 30.1% (incl. family trusts) and Randall Ruhlman with sole control of about 17.7% more, about 48% combined by arithmetic (proxy tables can double-count shared trusts).
6. **Reinvestment with healthy unit economics.** Unit economics = whether each sale or customer is profitable. Amazon went public in 1997 and first reported a full-year profit in 2003. Red flag: each customer also loses money.
7. **Customers who stay and spend more.** Churn = rate customers leave; net retention = what existing customers spend vs. last year. FICO's platform business: 148%.
8. **Survivable dilution and a cash runway.** Dilution = new shares shrink each existing slice. Runway = time before cash runs out. AST SpaceMobile's share count

rose about 18% in one year. Check the 5-year share count trend.

9. **Small starting size.** 68% of multibaggers in Martelli's study traded below \$300 million at their low.

Lesson 7: Reading a financial report

Public companies report every quarter. The money flows top to bottom:

1. **Revenue:** all money in from sales.
2. Minus cost of goods sold = **gross profit**.
3. Minus operating expenses = **operating income** (profit from running the business).
4. Minus interest and taxes = **net income** (the bottom line).
5. **EPS** = net income / shares: your slice of profit.

Free cash flow (FCF) = cash generated minus spending on buildings and equipment. Generally harder to manipulate than accounting profit, though not impossible, which is why your valuation method prefers it.

Lesson 8: Seven analyst tools

1. **Operating leverage.** Fixed costs stay flat, so extra sales are mostly profit. PLPC 2026: sales +25%, EPS +75%. In reverse, 2024: sales -11%, profit -41% (\$63M to \$37M). High-fixed-cost companies tend to boom and bust harder than their sales.
2. **Backlog and book-to-bill.** Backlog = signed, undelivered orders. Book-to-bill = new orders / shipments; above 1.0 the backlog grows. Hammond's backlog fell 6.9% in one quarter; management said higher production let it ship more (the company's explanation).
3. **Adjusted vs. reported earnings.** Reported (GAAP) follows official rules; adjusted removes costs the company calls one-time. Trust adjusted figures more when the gap shrinks over time.
4. **Three P/E lenses.** P/E = price / EPS (years of current profit you pay for). Trailing = last 12 months; forward = next year's estimate; run-rate = latest quarter x 4. PLPC: ~48x trailing vs. ~24x run-rate (calculations). Hammond: ~51x trailing vs. ~23x forward (data provider; forward relies on estimates).
5. **Organic vs. acquired growth.** American Superconductor's 30% growth was partly acquired. A widely cited HBR analysis (2011) put acquisitions failing to deliver expected value at 70-90%.
6. **Cyclical vs. structural.** Check 10+ years of revenue. PLPC grew about 4% a year from 2013 to 2025, with an 11% drop in 2024: a cyclical pattern despite a spectacular 2026.
7. **Share structure and control.** Dual-class shares give founders extra votes. As of September 2024, William Hammond owned about 26.9% of Hammond's shares but 57% of the votes (may have changed since). Also check which share classes a market cap includes: Hammond is about C\$2.4 billion (Class A only) or C\$3.2 billion (both), by calculation.

Lesson 9: Case studies

Company	What it looked like	What it taught
Powell	Data center switchgear boom, ~27% returns on capital (data provider)	A titan after discovery: much of the value growth has likely already happened
PLPC	Record profits, family-controlled, almost no debt	A cyclical at a possible peak can look like a titan; check long history
Hammond	Backlog +122% at end of 2025	Shortage booms can be

Harmonia	Backlog +122% at end of 2023, +96.9% as of Q2 2026	Shortage booms can be temporary unless the moat outlasts the shortage
American Superconductor	Strong headline growth	Growth partly bought, margins falling, shares diluting: three red flags
AST SpaceMobile	World-changing vision	In my view priced for a great deal of success; share count up about 18% in a year
AbCellera	Promising drug results	Value depends heavily on one drug program, ABCL635
Rocket Lab	Real business, strong growth	Too big for a realistic 100x; its early multiplier happened years ago

Lesson 10: Your personal edge

Philip Fisher called it **scuttlebutt**: learning about a business from the people who use its products, before it shows up in reports. Working in transmission and substation construction, you can observe:

- Which equipment brands utilities specify and crews prefer
- What's backordered or constantly delayed
- Whether project volume is growing or slowing
- Which contractors and suppliers are winning work

Many investors see this only in quarterly reports. Some analysts do their own industry checks, but your daily view of the job site is a genuine information source.

Cheat sheet

1. **Titans are rare:** about 10% of stocks were big winners (J.P. Morgan, 1980-2020); 4% created all net wealth (Bessembinder, 1926-2016).
2. **Losses are common:** about 40% fell 70%+ and never recovered; about 40% lost money outright.
3. **Winners fell hard too:** recent 10-baggers had a median worst decline of 56%; the 100 best stock-decades averaged declines over 30%.
4. **Size:** market cap = price x shares; 68% of multibaggers in Martelli's study were below \$300 million at their low.
5. **S-curve:** smartphones went from 35% to 81% of US adults in eight years.
6. **Second-order thinking:** find the bottleneck behind the obvious trend.
7. **Six boom signals:** big buyers commit, lead times stretch, costs fall, adoption is low, first pure-play profits, stocks haven't spiked.
8. **Nine traits:** wave, real accelerating product, moat, rising margins, owner-leader, reinvestment with good unit economics, sticky customers, little dilution, small size.
9. **Money flow:** revenue -> gross profit -> operating income -> net income -> EPS; free cash flow is generally hardest to manipulate.
10. **Seven tools:** operating leverage, backlog and book-to-bill, adjusted vs. reported, three P/E lenses, organic vs. acquired (70-90% of acquisitions fail to deliver), cyclical vs. structural, share structure and control.
11. **Survivorship bias:** falling far is common among both winners and losers.
12. **Your job is a research tool.**

Sources

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- 10-bagger drawdowns, MicroCap Club: <https://microcapclub.com/the-myth-and-math-of-10-baggers-what-investors-overlook/>
- Martelli size data, via Mayer's *100 Baggers*:
<https://www.scribd.com/document/331548041/100-Baggers-pdf>
- Mayer traits and 2009-2023 tenfold-stock study, Forbes:
https://www.forbes.com/sites/bill_stone/2026/06/13/how-to-find-multibagger-stocks-before-they-soar/
- Company figures (PLPC, Hammond, Powell, AMSC, AST SpaceMobile) come from company filings and releases gathered in earlier research.
- Approximate, from memory (worth double-checking): Bessembinder's 4% finding, Pew smartphone figures, IRENA solar figure, HBR acquisition range, Mayer's 365-stock sample, Amazon's first-profit year, historical drawdowns and starting sizes.